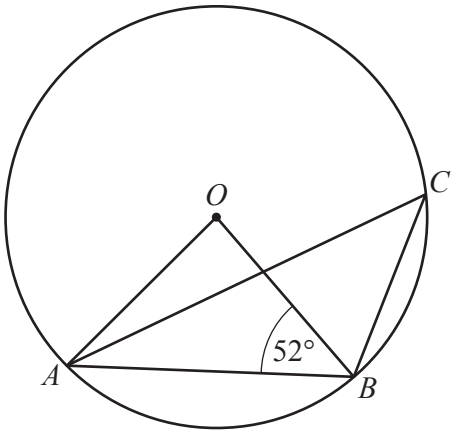
6 **(a)** The interior angle of a regular polygon is 156°.

Calculate the number of sides of this polygon.

..... [2]

(b)



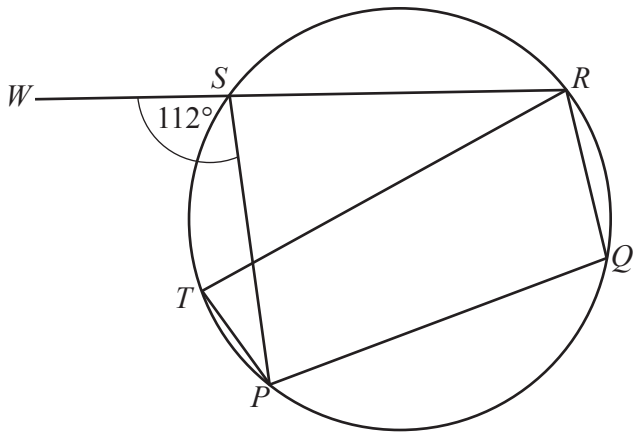
NOT TO SCALE

A, *B* and *C* lie on a circle, centre *O*.
Angle *OBA* = 52°.

Calculate angle *ACB*.

Angle *ACB* = ... [2]

(c)



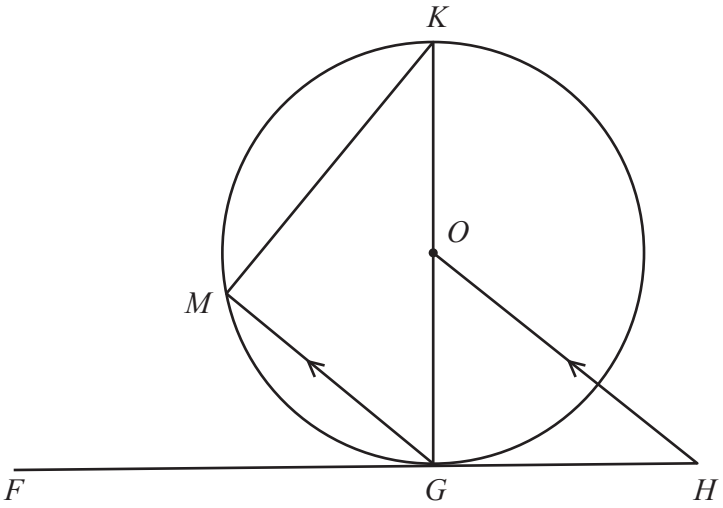
NOT TO SCALE

P, *Q*, *R*, *S* and *T* lie on a circle.
WSR is a straight line and angle *WSP* = 112°.

Calculate angle *PTR*.

Angle *PTR* = ... [2]

(d)



NOT TO SCALE

G, *K* and *M* lie on a circle, centre *O*.
FGH is a tangent to the circle at *G* and *MG* is parallel to *OH*.

Show that triangle *GKM* is mathematically similar to triangle *OHG*.
Give a geometrical reason for each statement you make.

.....

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..... [4]